

Bit wise shift

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$$-8 = 1000$$

2³ 2² 2¹ 2⁰ MSB

Make it 8 bit = 0000 1000

question is $-8 << 1$ (left shift by 1 bit)

MSB

$$0000 1000$$

MSB

$$0001 0000$$

2⁴ 2³ 2² 2¹ 2⁰

In - decimal no., the MSB
i.e. the left most bit is 1.

So we got answer as

$$1001 0000 = -16$$

2⁴ 2³ 2² 2¹ 2⁰

Short Method to get direct output

$$-8 << 1$$

If left shift then multi plⁿ
by no. of times the shifting occurs,

here by 1 time i.e. 1 time = 2 bit

$$\text{So } -8 \times 2 = -16$$

9/6 Right shift by 1 then
divide the given decimal by 2 to
get the answer.

$$-8 \gg 1 \quad (\text{Right shift by 1})$$

$$-8/2 = -4 \quad (\text{Short Method})$$

$$\text{Main 8 bit} = \textcircled{0}000\ 1000 \quad \downarrow$$

$$\gg 1$$

$$\textcircled{0}000\ 0100 = -4$$

$2^1\ 2^0$

Sign bit will be $\textcircled{1}$.